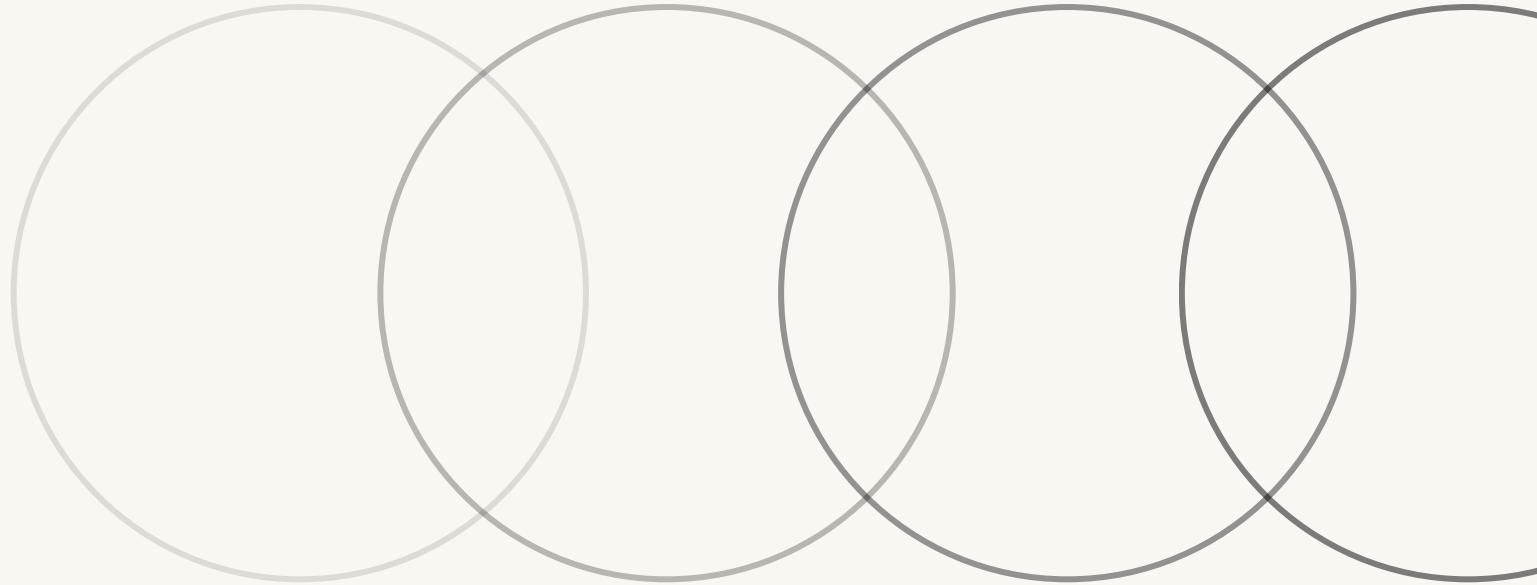
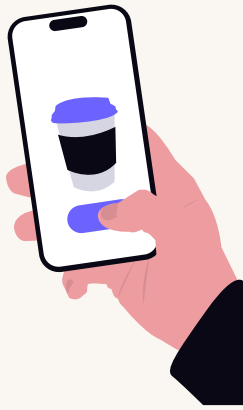




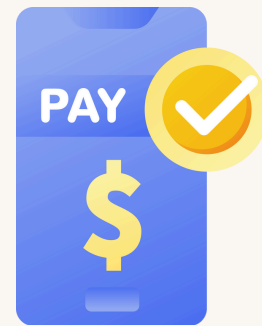
Presented By:

➤ Sabrina Heng Wei Qi ➤ Woo Cheng Shuan ➤ Lau Yan Kai ➤ Chew Chiu Xian ➤ Neo Li Xin

Action Flow (System Involved)



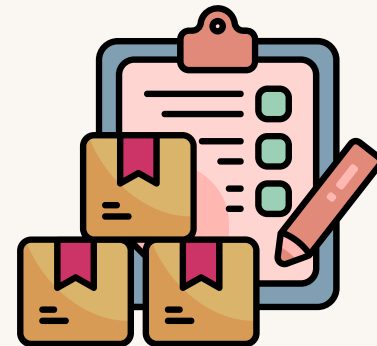
**Customer Mobile App /
Driver App**



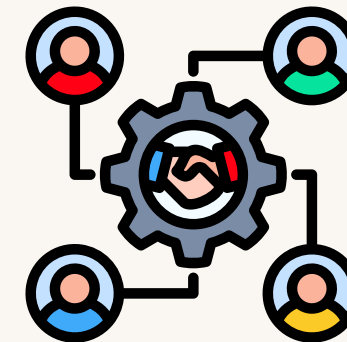
**Payment Gateway
System**



**Dispatch and Routing
System**



**Restaurant Inventory
Management System**

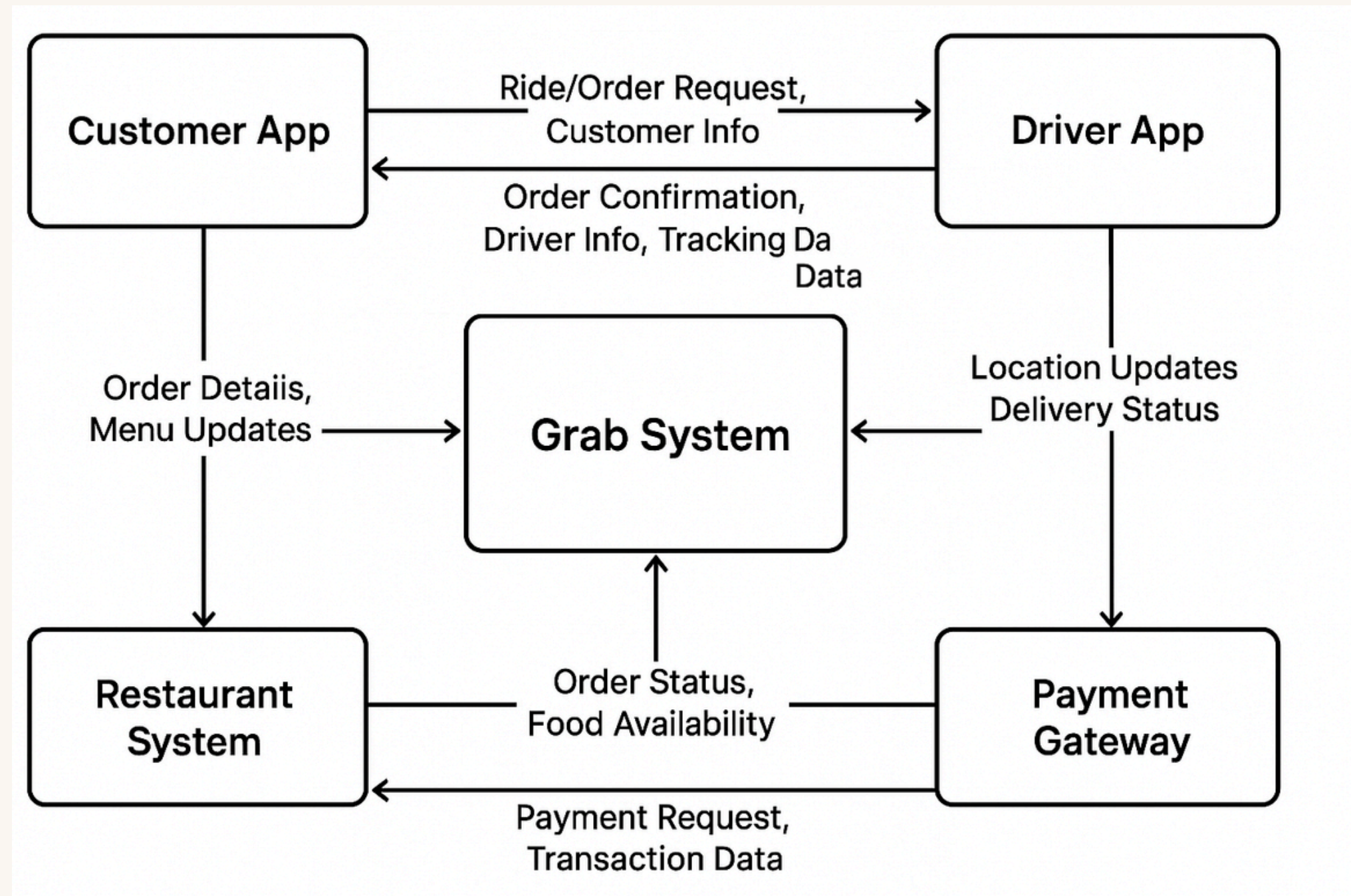


**Customer Relationship
Management (CRM)
System**



**Backend Server and
Database**

Action Flow (Data Exchange)

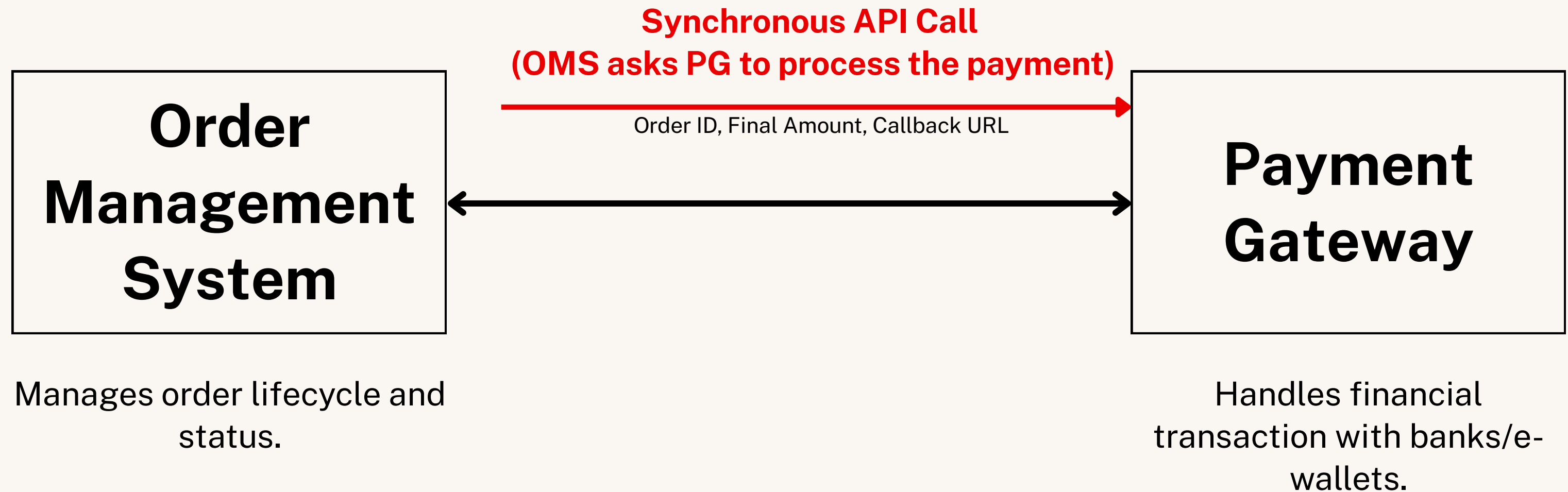


Grab App - Data Flow Diagram

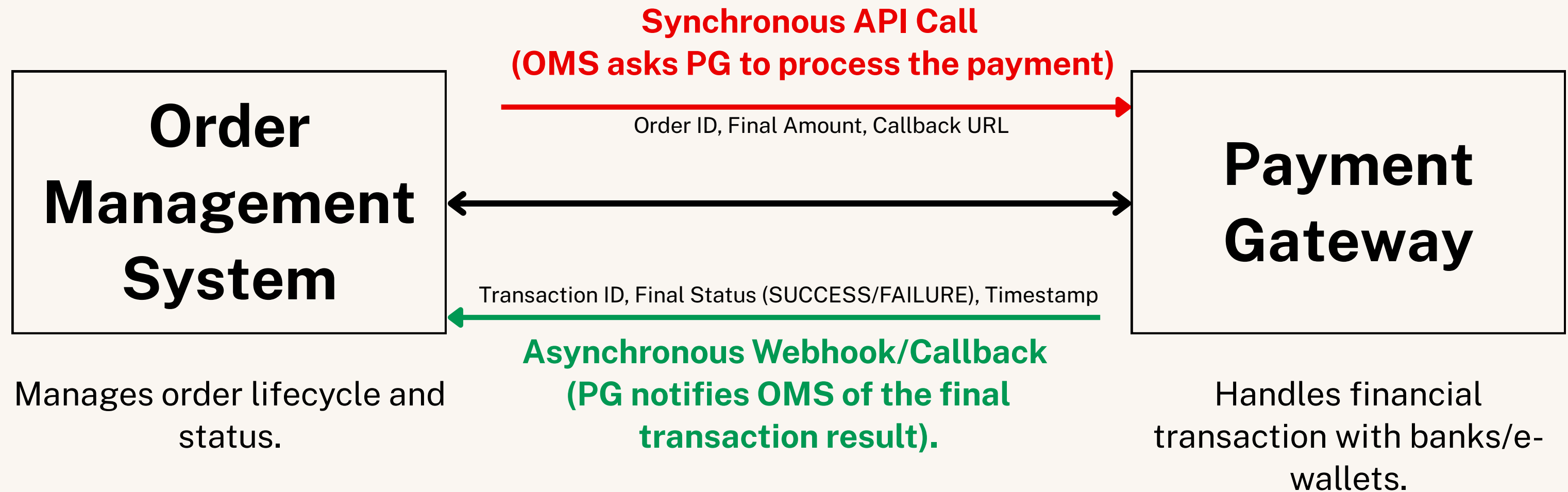
Integration Point

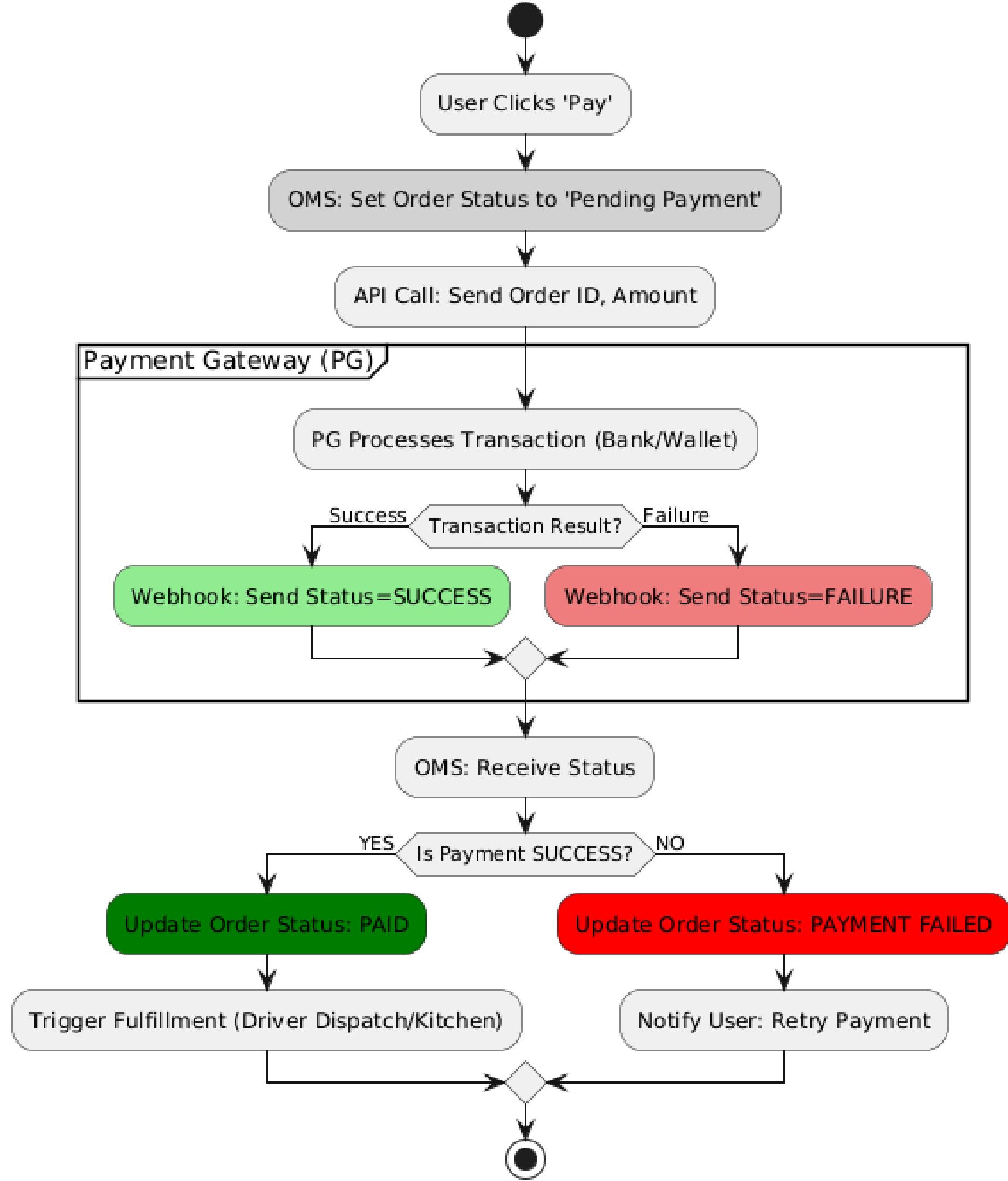


Integration Point

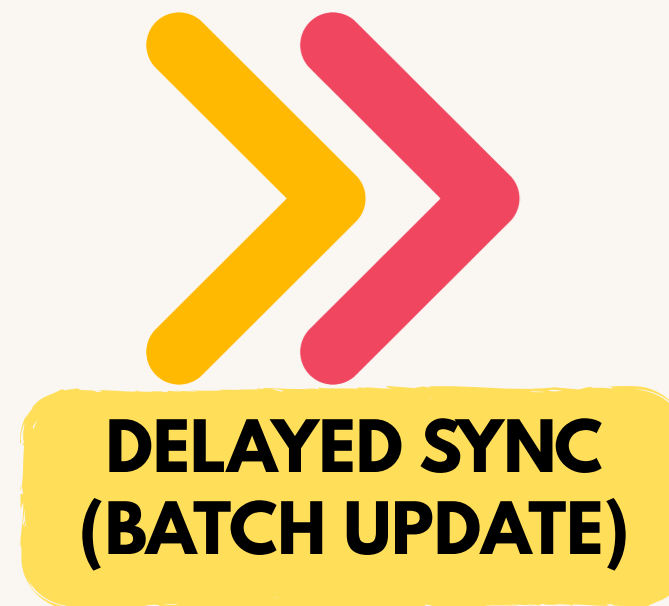


Integration Point





Logical Failure: The Inventory Mismatch



THE PROBLEM: ASYNC DATA

Local Inventory → Grab App
(30 min Sync)



IMPACT: CUSTOMER & OPS

Frustration, Cancellations,
Manual Support



CLASSIFICATION: LOGICAL ISSUE

Inconsistent Data (Design Flaw)

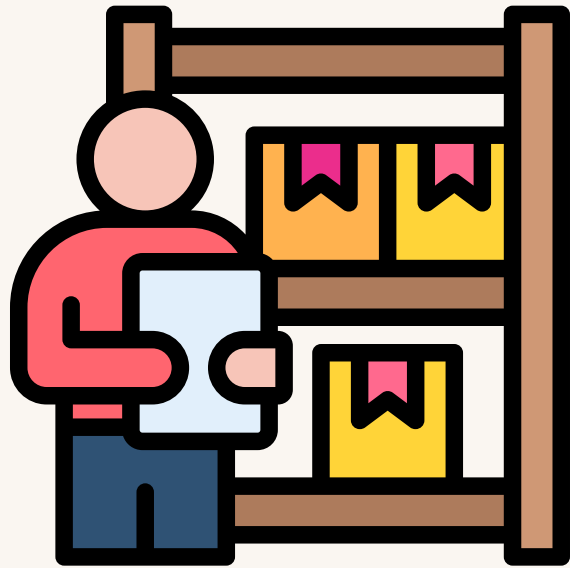
Physical Integration Failure



Failure Senarios	Impact to Users and Operations
Server downtime at Grab's backend	Customer cannot place or confirm ride/food orders
Network failure / API timeout between Grab app and payment gateway	Payment deducted but order not recorded → refund issues
GPS malfunction or signal loss affecting driver tracking	Driver cannot receive booking requests → idle time and income loss
Database crash causing booking records not be stored	Delayed or missing live tracking → customer anxiety and complaints
Push notification service failure (Firebase / APNs down)	Support team flooded with manual reports and disputes

Improvement:

Implement Real-Time Inventory Sync using APIs or Webhooks



- Replace batch updates with event-driven integration.
- Whenever a merchant marks an item as “Sold Out” in their POS system, a webhook instantly sends that update to the Grab backend.
- The Grab app refreshes the menu in real time, preventing customers from ordering unavailable items.

Benefits:

- No more “sold-out but still orderable” issues
- Fewer cancellations and refund requests
- Higher customer satisfaction
- More reliable and consistent data across systems

